

Motor Insurance claim frequency modelling

(Exposure-Adjusted Poisson GLM in R)

SUMMARY:

This project develops a claim frequency model using an exposure-adjusted Poisson GLM on a motor insurance dataset. The objective is to estimate expected claim counts across policyholders using demographic and vehicle risk factors. The results are used to derive risk insights and demonstrate how frequency modelling supports insurance pricing decisions.

PROBLEM:

Insurance companies must estimate how frequently claims are expected to occur across different policyholders. Since risk varies across customers, accurate claim frequency estimation is essential for fair pricing and portfolio risk management.

This project focuses on modelling expected claim counts based on observable risk characteristics.

DATA:

The dataset consists of over 678,000 motor insurance policies, including claim counts, exposure duration, driver age, vehicle characteristics, bonus-malus score, and urban density. The portfolio is largely composed of middle-aged drivers, with driver ages ranging from 18 to 100 years. The large sample size ensures statistical reliability for modelling claim frequency patterns.

The claim distribution is highly skewed, with most policies having zero claims, indicating a rare-event structure suitable for count-based modelling.

METHOD:

A Poisson Generalized Linear Model (GLM) with log link was used to model claim frequency.

The model assumes a multiplicative relationship between predictors and expected claim frequency.

Model:

$$\text{ClaimNb} \sim \text{DrivAge} + \text{VehAge} + \text{VehPower} + \text{BonusMalus} + \text{Density}$$

An offset term $\log(\text{Exposure})$ was included to account for varying policy durations, ensuring that claim frequency is measured per unit exposure.

INTERPRETATION:

Claim frequency varies significantly across policyholder segments.

Higher bonus-malus scores are associated with increased expected claims. Vehicle power shows a positive relationship with claim frequency, indicating higher risk for more powerful vehicles. Urban density also contributes positively to claim frequency, likely due to higher traffic exposure.

Vehicle age shows a negative relationship with claim frequency, suggesting relatively lower claim occurrence for older vehicles.

The model demonstrates that claim risk is not uniform across the portfolio and depends on multiple observable factors. These differences highlight the need for differentiated pricing and risk-based underwriting decisions.

These findings provide a basis for risk differentiation across policyholders.

Overall, the model shows that claim frequency increases with vehicle power, bonus-malus score, and urban density, while it decreases with vehicle age.

FURTHER ANALYSIS:

The dataset was also split into training and testing samples to evaluate model predictions out-of-sample.

A simple pricing illustration was performed by converting predicted claim frequency into premium estimates using an assumed average claim cost.

SUGGESTIONS:

Pricing should be adjusted based on risk levels observed in the model.

Segments with higher expected claim frequency, such as high vehicle power or high-density areas, can be charged relatively higher premiums.

At the same time, lower-risk customers can be offered more competitive pricing.

ASSUMPTIONS AND LIMITATIONS:

This analysis focuses only on claim frequency as claim severity data was not available.

A constant average claim amount was assumed for pricing, so the premium values are only indicative.

The Poisson model assumes that mean equals variance, which may not fully hold in this dataset.

Other models like Negative Binomial can be explored for better accuracy.

CONCLUSION:

This project establishes a baseline framework for estimating motor insurance claim frequency using exposure-adjusted GLM techniques. The results highlight key risk drivers and demonstrate how statistical modelling can support actuarial decision-making in pricing and risk assessment.